

Project Design Phase

Solution Architecture

Date: 19 February 2026

Team ID: LTVIP2026TMIDS72781

Project Name: Flight Booking App (MERN Stack)

Maximum Marks: 4 Marks

Architecture Overview (MERN Stack)

- User interacts through Web Browser (React.js Frontend).
- Frontend communicates with Backend (Node.js + Express.js REST APIs).
- Backend connects to MongoDB Atlas Database.
- External Integrations: Payment Gateway (Razorpay/Stripe) and Email Service (Nodemailer).
- Cloud Deployment on AWS / Render / Vercel.

Data Flow Summary

- 1. User registers/logins → JWT authentication issued.
- 2. User searches flights → Backend fetches data from MongoDB.
- 3. User books flight → Payment API processes transaction.
- 4. Payment success → Booking stored → Confirmation email sent.